Multi-Sensor Fusion for Remote Sensing of Metallic and Non-Metallic Object Classification in Complex Soil Environments and at Different Depths

Misha Urooj Khan, Muhammad Ahmad Kamran, Wajiha Rahim Khan, Malik Muhammad Ibrahim, Muhammad Umair Ali, and Seung Won Lee

***Abstract*— The global metal detector (MD) market is predicted to grow to U.S. $2.74 billion by 2030, with a compound annual growth rate (CAGR) of 7.66%, indicating potential expansion in the industry. Metallic and non-metallic material/object detection is pivotal to various technical applications and paramount in ensuring structural integrity, operational efficiency, and overall safety. This study analyzes a multi-sensor fusion approach for classification of metallic and non-metallic objects in complex soil environments at various depths. The adopted approach is beneficial in scenarios where single-sensor systems fail to provide adequate performance. We investigated the integration of magnetic induction spectroscopy (MIS) and ground penetrating radar (GPR) sensors for binary, soil, depth, and multiclass classification. A dataset comprising 14 objects was trained and tested on six AI models (900 variants). The best-performing 384 model variants, which have the highest accuracy, precision, and recall, reaching 98%+ and with minimal Type I and II errors, are reported here. The best accuracy for each classification problem (binary, soil type, depth, and multiobject) corresponding to each set of input data (MIS, GPR, and MIS+GPR) has been summarized and reported. The analysis shows that different AI models have performed better in different scenarios. The study's implications extend to defense, archeology, and geology, offering enhanced object detection in complex soil environments. These findings underscore the multi-sensor fusion approach's significance for advancements in remote sensing and classification technologies.**

***Index Terms*— Complex soil environments, ground penetrating radar (GPR), industry applications, landmine detection, magnetic induction spectroscopy (MIS), multi-sensor fusion, object classification, remote sensing.**

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I. Introduction

METALLIC and non-metallic object detection plays a pivotal role in various industries, ensuring diverse systems' integrity, safety, and functionality. Industry reports predict that the global metal detection market will reach U.S. $1.5 billion by 2025, owing to an increasing demand for safety and quality control [1]. The non-metallic detection market is projected to grow substantially, driven by the expansion of its food and pharmaceutical industries. For metallic objects, landmines play a critical role in the defense industry, serving as formidable warfare tools, yet their inherent danger is undeniable [2]. These mines are strategically deployed in both planned and unplanned operations. Over the past decade, mines buried in the grounds have penetrated deeply into the soil due to geological shifts and natural disasters, e.g., landslides, earthquakes, and floods, and with time, rainwater has carried them to unknown regions. Landmines profoundly impact the global defense landscape, with staggering statistics revealing their prevalence, as in Fig. 1.

It is alarming that these deadly devices can be produced for as little as U.S. $1, with prices varying between U.S. $3 and $75 per unit [3]. The scale of the issue is vast, with an estimated 10 million landmines scattered across 64 countries worldwide and an additional 2–5 million mines being laid each year [4]. On average, these indiscriminate killers claim the lives of 1000–2000 people yearly, primarily innocent civilians [4], [5]. The cost of clearance, estimated by the United Nations, including support and logistic expenses, is between U.S. $300 and $1000 per mine [5]. Efforts to mitigate the devastating consequences of landmines are imperative for the international community's security and humanitarian goals. Several classical techniques exist for buried metal/landmine detection. Nuclear detection techniques, which have been researched since the 1940s, include nuclear quadrupole resonance (NQR) and neutron-based methods [6]. Mechanical detection involves plows for unearthing and moving landmines, flails, and rollers for triggering and detonating them. Current automated technology is limited to manual probing. Using ultrasound and acoustic-to-seismic (A/S) techniques,

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acoustic detection is viable for metal detection and identification. Electromagnetic detection, based on disparities in electromagnetic properties, employs various techniques with distinct operating frequencies, electromagnetic spectrum bandwidths, transmitted signal types, signal interpretations, and transmitter–receiver types [7]. Standard methods include metal detectors (MDs), ground penetrating radar (GPR), microwave radar (MWR), millimeter-wave radar (MMWR), electrical impedance tomography (EIT), and infrared (IR) methods. Optical detection, limited by optical wavelength penetration, measures soil surface properties affected by buried landmines. Visible light and light detection and ranging (LiDAR) are standard optical detection techniques [8]. The performance of each technique depends on the soil type, environment, and metal and landmine type.

For metallic objects, landmine clearance remains a significant global challenge, impacting millions of lives. Recent advancements in detection technologies are focused on machine learning (ML) techniques applied to GPR [9], MDs [10], and prompt gamma neutron activation analysis (PGNAA) [11]. Núñez-Nieto et al. [12] assessed the efficacy of GPR equipped with 2.3- and 1-GHz high-frequency antennas with ML techniques for automatic detection of underground explosive landmines using neural networks and logistic regression (89%–92%). Safatly et al. [13] addressed the limitations of manual MD-based landmine clearance by ML. Convolutional neural networks (CNNs), XGBoost, and bagging classifiers are employed to discriminate landmines and buried objects with 90% accuracy. Baur et al. [14] used lightweight multispectral and thermal IR sensors for rapid landmine detection with 71.5%–98.3% accuracy. Tubaro et al. [15] proposed a CNN for landmine detection using GPR images with 95% detection accuracy on synthetically generated data. Xu et al. [16] investigated the performance of a PGNAA system for lower weight landmine detection using a neural network and XGBoost. Zhang et al. [17] proposed a software-defined Doppler radar (SDDR) system to distinguish non-metallic landmines from other buried objects, achieving 80% recall and 75% precision. Qiu et al. [18] proposed a UAV-based solution for scattered landmine detection via a multispectral fusion strategy. Baur et al. [19] used deep learning for scattered landmine detection with a region-based CNN (R-CNN), achieving a 91.8% detection rate for antipersonnel mines. Sun et al. [20] proposed landmine detection using forward-looking GPR and an adaptive learning approach. The significant limitations described in the literature are mentioned in the following.

1) Detection accuracy variations across soil conditions, challenge software-defined radio-based radar (SDRR) systems.
2) Traditional ML classifiers and CNNs require significant data to detect small objects (landmines).
3) Discrepancies between simulated and real data collection and minefield environments impact the reliability of detection methodologies.
4) Fusing unpiloted aerial systems and deep learning requires optimization for complex real-world scenarios.

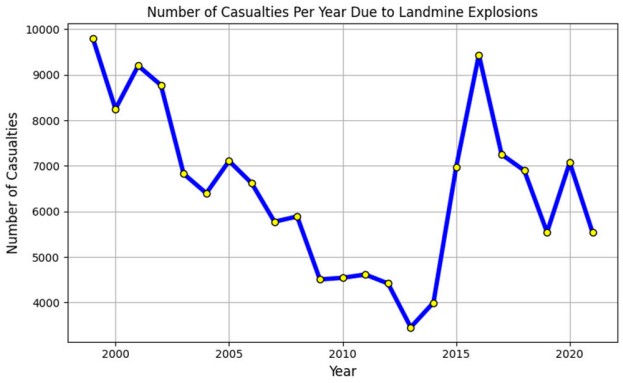


Fig. 1. Casualties caused by global landmine explosions.

5) Challenges in precision–recall balance, dataset completeness, clutter handling, feature set optimization, and adaptability to diverse conditions persist in almost all metallic (landmine) and non-metallic detection technologies.

Thus, these studies collectively emphasize the need for extensive datasets, innovative fusion strategies, and adaptive learning approaches to increase the efficacy of object detection systems. The practical implementation of the proposed methodology in Fig. 2 will significantly enhance the remote sensing capabilities of GPR and MIS for object detection purposes. The primary contributions and steps of this study are outlined as follows.

1) Detailed experimentation is performed using the MIS/GPR dataset [21], [24]:
   a) binary object classification (metallic versus non-metallic);
   b) multiple soil classifications (cooperative, uncooperative, and rocky);
   c) multidepth classification (5 cm, 10 cm, and flush);
   d) multiobject classification (14): metallic (landmine—PMA2 & 3, bullet (55 and 88 mm), coin (1 and 10 euro), mortar bomb, ball bearing, and ring pull) versus non-metallic (rock—1 and 2, and wood—S&M).
2) Multiple techniques, such as data transformation, segregation, and preprocessing, are applied to the raw dataset. GPR preprocessing involves dc subtraction, segmentation, empirical mode decomposition (EMD), a dewow filter, and data augmentation to address class imbalance. MIS signal preprocessing includes tuple conversion, absolute value conversion, moving average filtering, segmentation, EMD, and data augmentation.
3) Sixty-six features are extracted from six domains, and then, standardization is applied, followed by dataset splitting into training and test sets.
4) Six models, namely, support vector machine (SVM), k-nearest neighbor (KNN), decision tree (DT), random forest (RF), multilayer perceptron (MLP), and 1-D CNN (1DCNN), are trained and tested with more than 900 variants.

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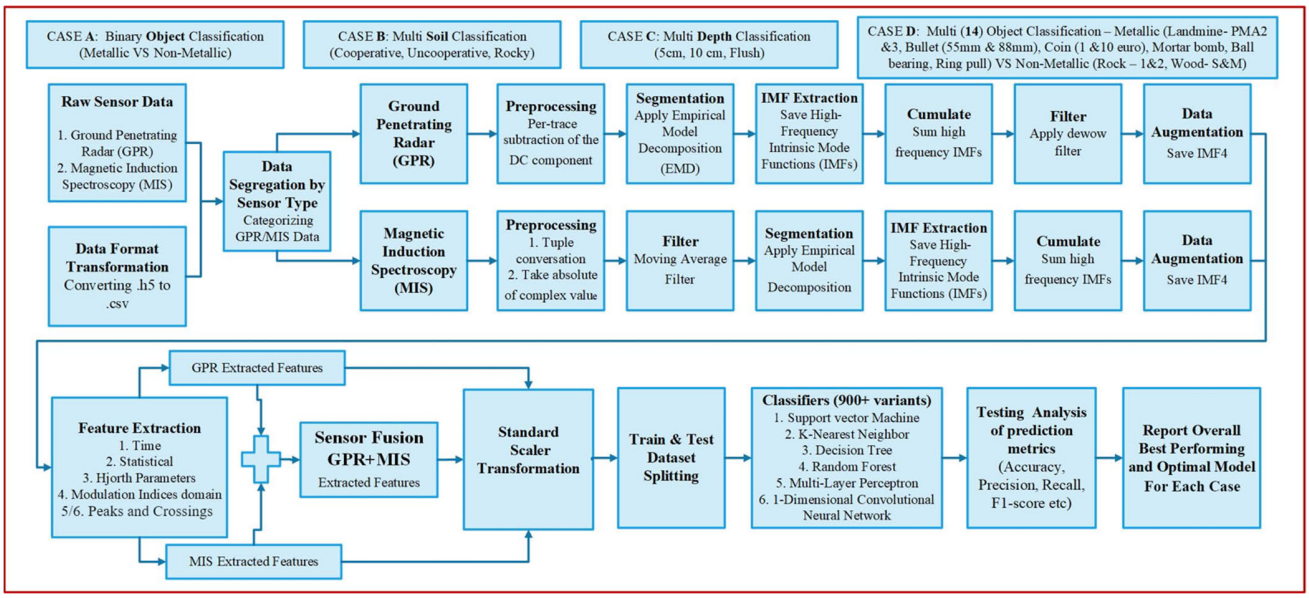


Fig. 2. Flow diagram of the detailed methodology.

5) For performance evaluation, the prediction metrics (accuracy, precision, recall, and F1-score) are analyzed in detail.
6) After extensive analysis of the results, the best-performing model for each case (binary, soil, depth, and multiobject classification) is separately reported.
7) Finally, a comprehensive model comparison summary encompassing various parameters, such as weight, k (number of neighbors), criterion, number of trees, kernels, hidden layer size, batch size, learning rate, and optimizer, is provided, focusing on training time and feature requirements (global/local features).

II. Materials and Methods

A. Dataset

The dataset utilized in this study was collected using a dual-modality detector system (DMDS) and a deployable multifrequency magnetic induction spectroscopy (MIS) sensor system [21], as shown in Fig. 3, specially designed for remote sensing purposes. The dataset was collected in May 2018 at an experimentation site located in Benkovac, Croatia, at an HCR-CTRO-managed test site [22], [23]. The spectroscopic responses of the objects were measured in the air after they were buried at various depths (flush, 5 cm, and 10 cm) in unprepared, naturally rough soil. The test site had three soil types: homogeneous mineralized (uncooperative), homogeneous nonmineralized (cooperative), and heterogeneous mineralized (rocky). This dataset is sourced from [21] and [24]. Fig. 4 shows the time-domain plot, which features the raw GPR and MIS signals.

B. Preprocessing

The ".h5" files were converted to ".csv" files using a specialized parser (step 1), and the GPR and MIS data were separated into separate folders for a structured layout (step 2).

The raw dataset statistics in Table I(a) reveal the signal characteristics.

1) GPR Data Preprocessing Steps: To address challenges in GPR data for object detection, dc component subtraction is first used for centering the GPR signal through mean subtraction [25]. This step eliminates baseline drift, ensuring an accurate representation of signal variations specific to GPR data, thus mitigating amplitude variations and temporal distortions caused by heterogeneous subsurface materials

$$x_{\mathrm{ac}(t)} = x(t) - x_{\mathrm{dc}(t)} \tag{1}$$

where $x_{\mathrm{ac}(t)}$ represents the ac component of the signal.

$x(t)$ is the original signal. $x_{\mathrm{dc}(t)}$ is the dc component (mean) of the signal. Then, the EMD algorithm is applied iteratively for the extraction of intrinsic mode functions (IMFs) [26]. After analyzing random 50.csv files with their time and spectrogram visualization, it was found that the first three IMFs carried the most prominent high-energy frequency components and feature representations, which is why their cumulative sum was taken [27]. Then, a dewow filter was applied to eliminate any remaining low-frequency noise and wow artifacts from the GPR signal [28]. Mathematically,

$$H(s) = \frac{1}{\sqrt{\left\{1 + \left(\frac{s}{\omega_c}\right)^{2n}\right\}}} \tag{2}$$

where s is the complex frequency variable, ω_c is the cutoff frequency, and n is the order of the filter. We applied a fourth-order Butterworth dewow filter.

2) MIS Preprocessing Steps: To improve MIS data reliability, complex values are first converted to tuples, and then, their absolute values are taken. Then, a 20-point moving average filter is applied to mitigate noise and accentuate the underlying

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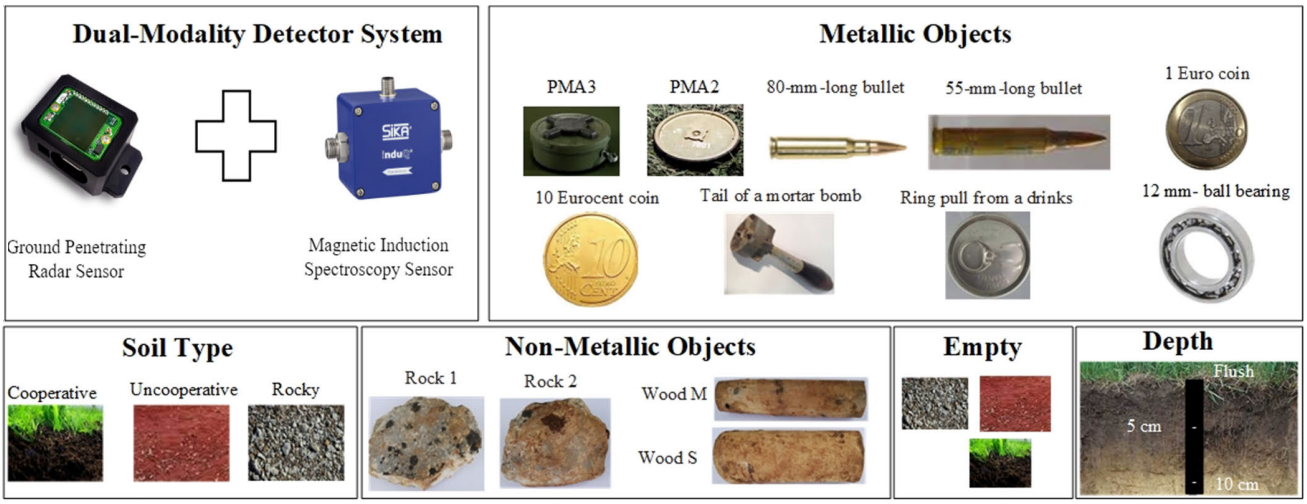


Fig. 3. Dual model detector system and objects used for data collection.

TABLE I

DATASET STATISTICS AFTER (a) FILE CONVERSION AND SENSOR DATA SEPARATION AND (b) PREPROCESSING

Sensor		(a)		(b)	
		.csv files	Samples per .csv file	.csv files	Samples per .csv file
MIS Position (cm)	x	1669	2422	2,971,539	1,980,347
	y	1669	5005	5,939,227	1,980,347
	z	1669	4991	5,926,249	1,980,347
GPR Frequency (kHz)	4.77	1669	5007	5,921,013	1,980,347
	12.40	1669	5007	5,921,013	1,980,347
	18.12	1669	5007	5,921,013	1,980,347
	28.61	1669	5007	5,921,013	1,980,347
	9.59	1669	5007	5,921,012	1,980,347
Total		**13,352**	**37453**	**44442079**	**15842776**

sharp step response in the MIS signals [29]. Mathematically,

$$\mathrm{FS}_i = \frac{1}{20} \sum_{j=i-10}^{i+9} \mathrm{MIS}[j]. \tag{3}$$

This equation represents the average of the 20 data points centered around the current index i in the "MIS" array, providing a smoothed representation of the original signal. EMD is applied to decompose the filtered signal into IMFs [30]. Then, the first three IMFs are summed to ensure the filtered sum signal, which accurately captures the most relevant information. The raw dataset statistics after all the preprocessing steps are shown in Table I(b).

C. Feature Extraction

Feature extraction reduces the dimensionality of the data and enhances interpretability [27], [31], [32], [33], [34], [35], [36], [37], [38]. Sixty-six features from six domains were extracted to train the AI models. In the statistics domain, the extracted features include the mean, standard deviation, variance, skewness, kurtosis, maximum, minimum, range, root mean square, energy, and mean absolute deviation [27]. In the time domain [33], the zero-crossing rate, entropy, autocorrelation, cross correlation, crest factor, form factor, peak-to-average ratio, signal-to-noise ratio (SNR), signal-to-distortion ratio, time-domain aliasing cancellation ratio, time-domain intermodulation distortion, time-domain total harmonic distortion, kurtogram peak, time-domain covariance, and autocorrelation lag are considered. Hjorth activity, Hjorth mobility, and Hjorth complexity were extracted for the Hjorth parameters. The amplitude and frequency modulation indexes were extracted in the modulation indices domain [33]. In the peak and crossing domains, the number of peaks, mean crossing rate, signal bandwidth, signal duration, signal crest factor, time-domain waveform length, time-domain zero crossings, and peak count were extracted [32]. The characteristics of a waveform or pulse can be analyzed by extracting various parameters such as the number of rising and falling edges, signal rise and fall times, signal symmetry, signal pulsewidth, signal impulse factor, signal clearance factor, signal overshoot, and signal undershoot [31]. Time-domain features provide insights into signals' temporal patterns and unpredictability [33]. Frequency-domain features reveal frequency distribution and strength [33]. Statistical features capture statistical properties that define the shape and variability of the signals [27]. Modulation indices quantify the signal's modulation extent [32] with information about signal amplitude and frequency variations over time [31]. Peak-related features highlight significant points in

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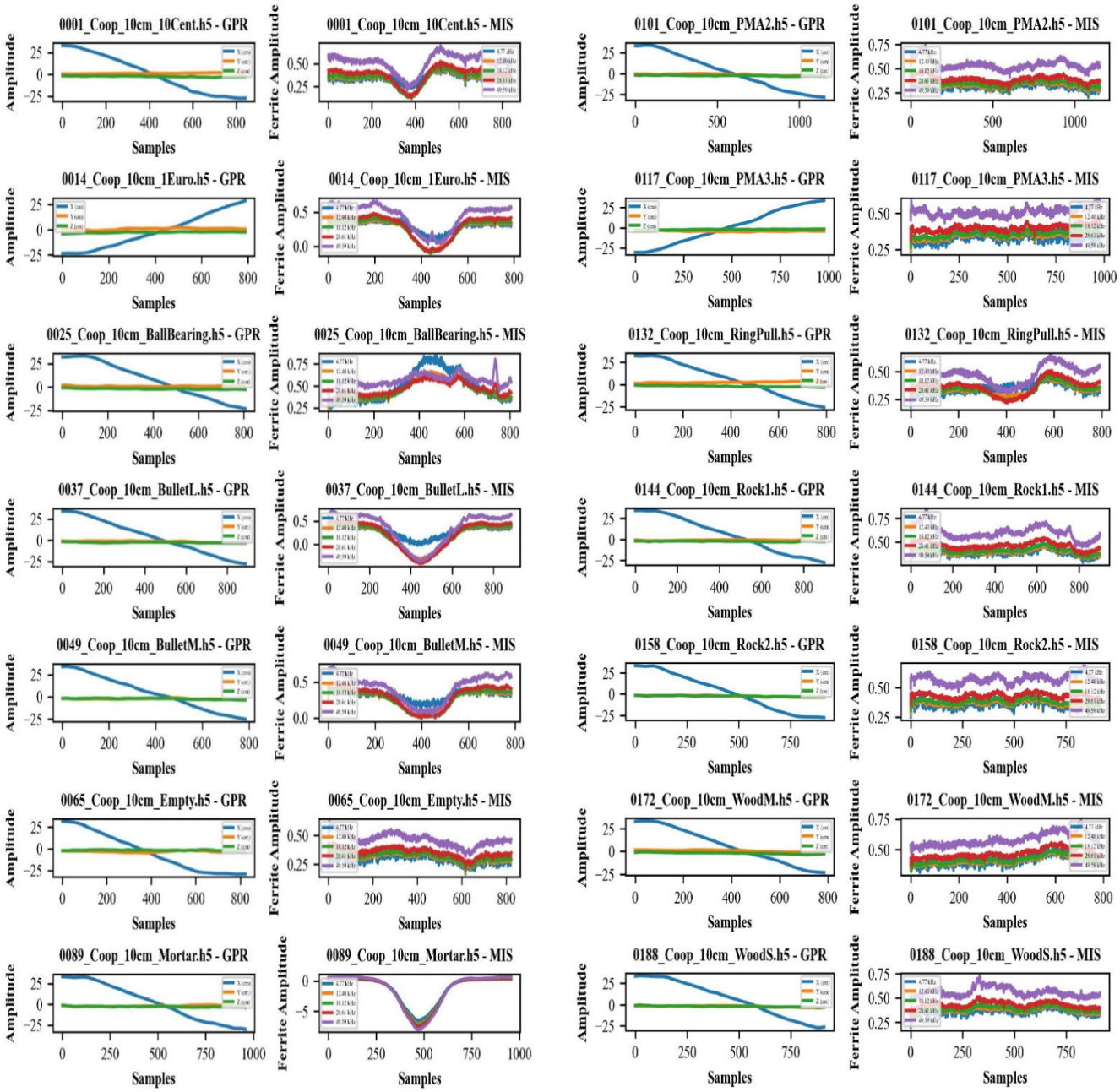


Fig. 4. Time-domain plot of raw GPR and MIS signals.

the signal, while crossings provide information about signal changes [27].

D. AI Models

Six AI models were trained and tested on the following extracted features: KNN, DT, RF, SVM, MLP, and 1DCNN. A total of 900 variants of six AI models were trained and tested.

However, only the best 384 variants are discussed here for analysis of the results.

1) *KNN:* In this study, KNN [27], [32], [33], [34] was evaluated for uniform and distance weights with one, three, five, and seven neighbors for all four classification cases. The total number of trained models comprised eight variants per signal, resulting in 24 variants per case and 96 trained variants of KNN.
2) *DT:* It was evaluated for the Gini and entropy criteria for all the classification cases [39]. The total trained models included two variants per signal, resulting in six variants per case and 24 trained variants of DTs.
3) *RF:* It was evaluated for the Gini and entropy criteria for MIS, GPR, and MIS+GPR features for all four cases [40]. The total number of trained models included two variants per signal, resulting in six variants per case and 24 trained RF variants.
4) *SVM:* Here, SVM [31], [35] was evaluated for linear, poly, RBF, and sigmoid kernels for each MIS, GPR, and

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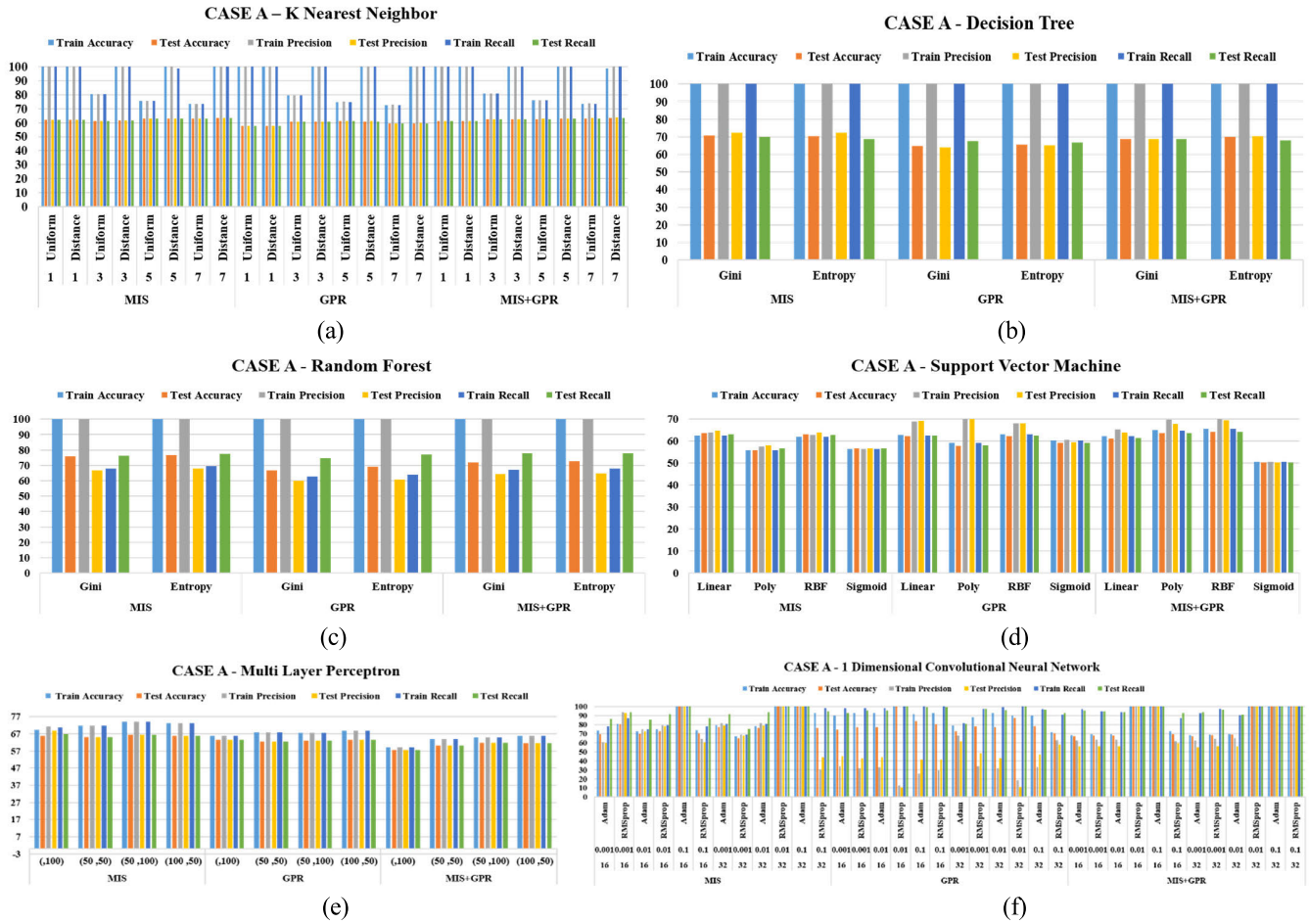


Fig. 5. Binary classification results. (a) KNN. (b) Decision tree. (c) Random forest. (d) SVM. (e) MLP. (f) IDCNN.

MIS+GPR feature. The total number of trained models included four variants per signal, resulting in a total of 12 variants per case and a total of 48 trained variants of SVM.

5) *MLP:* For this study, four different variants were evaluated for MIS, GPR, and MIS+GPR: (100) (50, 50), (100, 50), and (50, 100) for all four cases. The total number of trained models comprised four variants per signal, resulting in a total of 12 variants per case and 48 trained variants of the MLP [36], [37], [41].
6) *1DCNN:* For this study, the IDCNN has a 1-D convolutional layer with 64 filters, a kernel size of three, and rectified linear unit (ReLU) activation [42], [43]. Then, a MaxPooling 1-D layer with a pool size of two was added, and a flattened layer is to transform the output into a 1-D array, facilitating connection to densely connected layers [44]. An output layer with softmax activation is followed by two dense layers with 50 ReLU neurons. The model was compiled using categorical cross entropy as the loss function and an optimizer with a learning rate. The 1DCNN was evaluated for all four cases for MIS-, GPR-, and MIS+GPR-extracted features. The following configurations were evaluated for each case using different batch sizes (BSs), learning rates (LRs), and optimizers (Ops).
   a) BS: 16, LR: 0.001, and Op: Adam.
   b) BS: 16, LR: 0.001, and Op: RMSprop.
   c) BS: 16, LR: 0.01, and Op: Adam.
   d) BS: 16, LR: 0.01, and Op: RMSprop.
   e) BS: 16, LR: 0.1, and Op: Adam.
   f) BS: 16, LR: 0.1, and Op: RMSprop.
   g) BS: 32, LR: 0.001, and Op: Adam.
   h) BS: 32, LR: 0.001, and Op: RMSprop.
   i) BS: 32, LR: 0.01, and Op: Adam.
   j) BS: 32, LR: 0.01, and Op: RMSprop.
   k) BS: 32, LR: 0.1, and Op: Adam.

The total number of trained models included 12 variants per signal (GPR/MIS/MIS+GPR), resulting in a total of 36 variants per case and a total of 144 trained variants of 1DCNN overall.

III. Results and Discussion

The experimentation was conducted in Python 3.9 on a Windows OS, powered by an Intel Core i9-9700F CPU running at 3.00 GHz, supported by 16 GB of RAM, and a 2X RTX 3090 GPU with 48 GB of RAM for enhanced performance. The training test split ratio for the results mentioned from A to D included 80% of the samples in the training set, and the remaining 20% were unseen in the test dataset.

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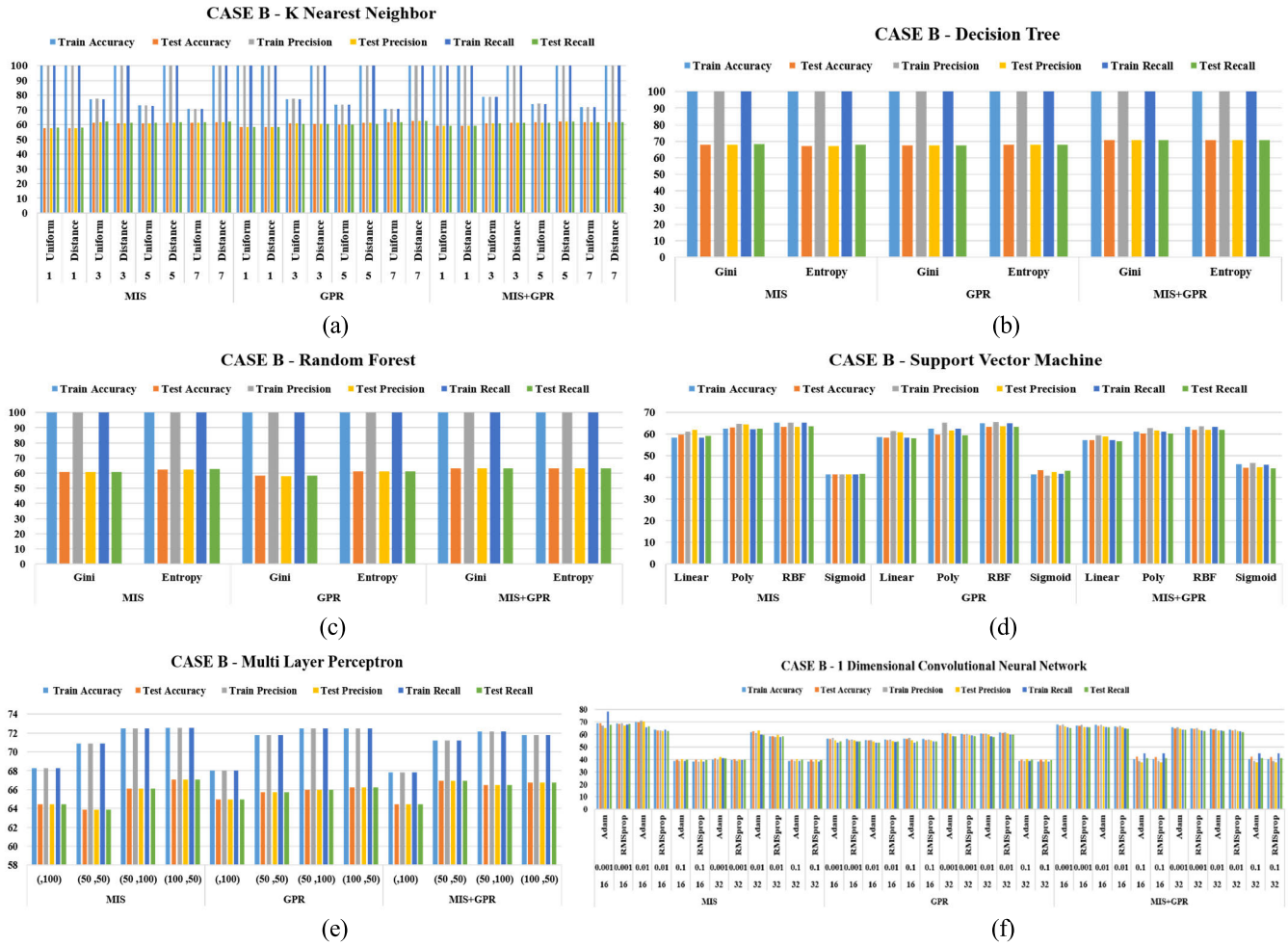


Fig. 6. Soil classification results. (a) KNN. (b) Decision tree. (c) Random forest. (d) SVM. (e) MLP. (f) IDCNN.

In Table II, we report the best-case scenarios of the test and training results for all cases. The model's overall performance is judged through only unseen (test) data; therefore, parameters, such as test accuracy, test precision, and test recall, were taken as evaluating parameters. We have evaluated the performance measure as the average of these three parameters, as shown in (4), but recall is a very sensitive parameter in this case. Therefore, one could also use a weighted average of these parameters. For all four cases (binary, soil, depth, and multiclass), the best model is declared the one with the highest % as

$$\text{Performance Metric(PM)} = \frac{1 * \text{Test accuracy} + 1 * \text{Test prescion} + 1 * \text{Test Recall}}{3}. \tag{4}$$

A. Object (Binary) Classification (Metallic Versus Non-Metallic)

MIS+GPR-KNN (7-distance) achieved an impressive 99% training accuracy, with test accuracy at 63.38%, test precision of 63.61%, a test recall of 63.39%, and an overall performance of 63.46%, as shown in Fig. 5(a). MIS-DT (Gini) achieved 99.99% training accuracy with 70.72% test accuracy and test precision and test recall are 72.36% and 69.99%, respectively, with an overall performance score of 71.03%, as shown in Fig. 5(b). MIS-RF (Entropy) reached 99.99% training accuracy with a test accuracy of 76.83%, a test precision of 67.83%, a test recall of 77.41%, and an overall performance of 74.03%, as shown in Fig. 5(c). MIS+GPR-SVM (RBF) achieved a training accuracy of 65.73% and a test accuracy of 64.23%, with a test precision of 69.64%, a test recall of 64.30%, with an overall performance of 66.05%, as shown in Fig. 5(d). MIS-MLP (100) resulted in a training accuracy of 69.3% and a test accuracy of 65.9%, with a test precision of 68.9%, a test recall of 67.1%, and an overall performance of 67.3%, as shown in Fig. 5(e). Among all, IDCNN-MIS (32, 0.01, RMSprop) and (32, 0.1, Adam), and IDCNN-MIS+GPR (16, 0.1, Adam), (32, 0.1, Adam), (16, 0.01, RMSprop), and (32, 0.01, RMSprop) stand out as the best with 99.99%, as shown in Fig. 5(f). It achieves the highest test accuracy and performance metrics, making it the most reliable model for deployment in real tasks. Table II(a) summarizes the performance rankings of the models with respect to each sensor for binary classification.

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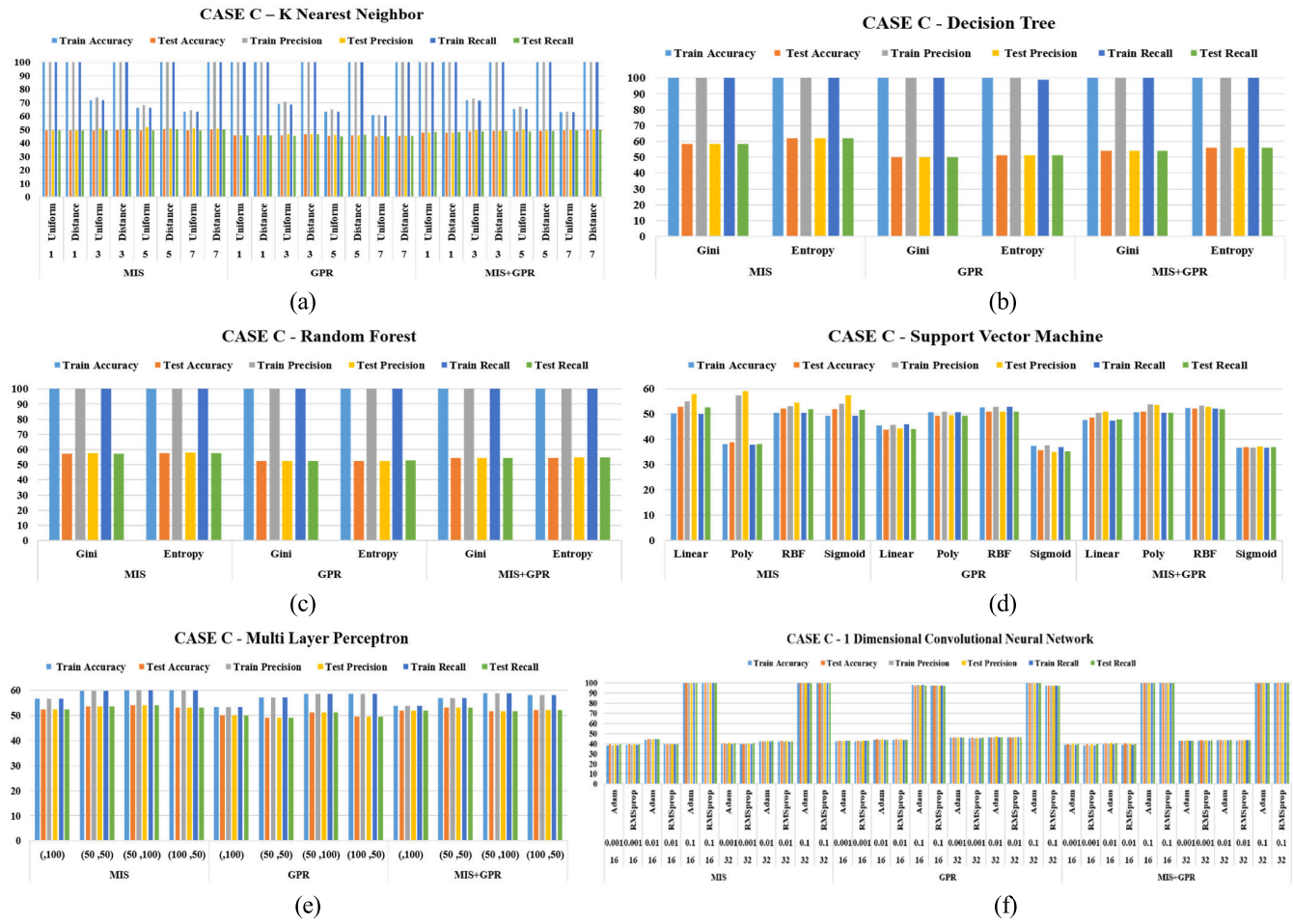


Fig. 7. Depth classification results. (a) KNN. (b) Decision tree. (c) Random forest. (d) SVM. (e) MLP. (f) IDCNN.

B. Soil (Multi) Classification (Cooperative, Uncooperative, and Rocky)

GPR-KNN (7-distance) achieved remarkable results, boasting a training accuracy of 99.99% with test accuracy of 62.37%, and has the test precision and test recall of 62.38% and 62.36%, respectively, resulting in an overall performance score of 62.37% [shown in Fig. 6(a)]. MIS+GPR-DT model (Gini) attained a 99.99% training accuracy, with test accuracy of 70.90% with test precision of 70.78% and test recall of 70.80%, culminating in an overall performance score of 70.82% [shown in Fig. 6(b)]. MIS+GPR-RF (Entropy) achieved a 99.99% training accuracy, with a test accuracy of 63.16%. Its test precision and test recall are at 63.04% and 63.09%, respectively, highlighting its robust performance, yielding an overall performance score of 63.10% [shown in Fig. 6(c)]. GPR-SVM (RBF) recorded a training accuracy of 65.17%, with a test accuracy of 63.52%, with a test precision of 63.61% and a test recall of 63.46%, contributing to an overall performance score of 63.53% [shown in Fig. 6(d)]. MIS-MLP model (**100**, **50**) demonstrated a training accuracy of 72.56%, with a test accuracy of 67.05%, with its test precision, test recall, and overall performance all coinciding at 67.05% [shown in Fig. 6(e)]. MIS-IDCNN characterized by (16, 0.01, Adam), with 70.22% training accuracy and 69.5% test accuracy with 70.3% test precision, 66.48% test recall, and overall performance at 68.76% [shown in Fig. 6(f)]. MIS+GPR-DT emerged as the frontrunner for real-time soil classification, and it solidified its position as the most dependable model for real-world deployment. Table II(b) summarizes the performance rankings of the models with respect t each sensor for soil classification.

C. Depth (Multi) Classification (5 cm, 10 cm, and Flush)

MIS-KNN (5-distance) boasted an impressive 99.99% training accuracy, with a test accuracy of 50.28%. Its test precision and test recall were at 50.76% and 50.38%, respectively, with an overall performance of 50.47% [shown in Fig. 7(a)]. MIS-DT model (Entropy) was achieved with a training accuracy of 99.99% and a test accuracy of 61.98%, with both test precision and test recall at 62.04%, with an overall performance score of 62.02% [shown in Fig. 7(b)]. MIS-RF (Entropy) reached a training accuracy of 99.99%, with a test accuracy of 57.67% with its test precision, test recall, and performance being 58.09%, 57.74%, and 57.83%, respectively [shown in Fig. 7(c)]. MIS-SVM (Linear) with a training accuracy of 50.29% and a test accuracy of 52.83%, with a test precision of 58.03%, a test recall of 52.62%, and an overall performance of 54.50% [shown in Fig. 7(d)].

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TABLE II
MODEL PERFORMANCE RANKING WITH RESPECT TO TRAIN AND TEST RESULTS FOR ALL CASES. (a) BINARY CLASSIFICATION, (b) SOIL CLASSIFICATION, (c) DEPTH CLASSIFICATION, AND (d) MULTI-CLASSIFICATION

(a)		
Sensor	**Train Results (%)**	**Test Results (%)**
MIS	KNN (99.99) = DT (99.99) = RF (99.99) = 1DCNN (99.99) > MLP (69.3) > SVM (62.56)	1DCNN (99.99) > RF (76.83) > DT (70.72) > MLP (65.9) > SVM (63.63) > KNN (63.43)
GPR	KNN (99.99) = DT (99.99) = RF (99.99) = 1DCNN (99.99) > MLP (68.84) > SVM (62.71)	1DCNN (99.99) > RF (69.23) > DT (65.71) > MLP (63.46) > SVM (62.36) > KNN (60.86)
MIS+ GPR	DT (99.99) = RF (99.99) = 1DCNN (99.99) > KNN (99) > MLP (65.78) > SVM (65.73)	1DCNN (99.99) > RF (72.71) > DT (69.78) > SVM (64.23) >KNN (63.38) > MLP (61.49)
(b)		
Sensor	**Train Results (%)**	**Test Results (%)**
MIS	KNN (99.99) = DT (99.99) = RF (99.99) > MLP (72.56)> IDCNN (70.22) > SVM (65.25)	IDCNN(69.5) > DT(67.91) > MLP(67.05) > SVM(63.38)> RF(62.45) > KNN (61.73)
GPR	KNN (99.99) = DT (99.99) = RF (99.99) > MLP (71.77) > SVM (65.17) > IDCNN (61.53)	DT (67.91) > MLP (65.76)> SVM (63.52) > KNN (62.37) > RF (61.29)> IDCNN (61.2)
MIS+ GPR	KNN (99.99) = DT (99.99) = RF (99.99) > MLP (71.20) > IDCNN (67.55) > SVM (63.52)	DT (70.90) > IDCNN (67.02) > MLP (66.97) > RF (63.16)> KNN (62.21) > SVM (62.01)
(c)		
Sensor	**Train Results (%)**	**Test Results (%)**
MIS	KNN (99.99) = DT (99.99) = RF (99.99) > IDCNN (99.88)>MLP (63.12) > SVM (50.29)	IDCNN (99.89) > DT (61.98)> RF (57.67) > MLP (53.99) > SVM (52.83) > KNN (50.28)
GPR	KNN (99.99) = DT (99.99) RF (99.99) > IDCNN (99.89) > MLP (58.55) > SVM (52.71)	IDCNN (99.89) > RF (52.59) > MLP (51.22) = DT (51.22) > SVM (50.93) > KNN (46.55)
MIS+ GPR	KNN (99.99) = DT (99.99) = RF (99.99) > IDCNN (99.89) > MLP (56.92) > SVM (52.36)	IDCNN (99.89) > DT (56.08) > RF (54.58) > MLP (53.17) > SVM (52.27) > KNN (49.86)
(d)		
Sensor	**Train Results (%)**	**Test Results (%)**
MIS	KNN (99.99) = DT (99.99) = RF (99.99) > IDCNN (19.24) > MLP (17.42) > SVM (13.35)	DT (47.62) > IDCNN (19.04) > RF (16.40) > SVM (12.23) > MLP (11.8) > KNN (10.65)
GPR	KNN (99.99) = DT (99.99) RF (99.99) > MLP (19.5) > SVM (13.32) > IDCNN (13.28)	DT (48.92) > RF (17.70) > SVM (14.96) > IDCNN (13.81) > MLP (12.81) > KNN (10.93)
MIS+ GPR	KNN (99.99) = DT (99.99) = RF (99.99) > MLP (23.01) > IDCNN (15.15) > SVM (15.31)	DT (46.13) > RF (23.49) > KNN (17.28) > 1DCNN (16.74) > MLP (16.56) > SVM(14.42)

MIS-MLP (**50**, **100**) with a training accuracy of 63.12% and a test accuracy of 53.99% with precision, recall, and overall performance, all equaling 53.99% [shown in Fig. 7(e)]. MIS-IDCNN (16, 0.1, Adam), (16, 0.1, RMSprop), (32, 0.1, Adam), and (32, 0.1, RMSprop), GPR-IDCNN characterized by (32, 0.1, Adam), and MIS+GPR-IDCNN (32, 0.1, Adam) and (32, 0.1, RMSprop) configurations emerged as the dominant player in this classification task. It achieved a training and testing accuracy of 99.89% with the highest test precision, test recall, and overall performance scores at 99.89% [shown in Fig. 7(f)]. Thus, IDCNN emerged as the best-performing model, showcasing the highest test accuracy and performance metrics, thereby solidifying its position as the most reliable choice for real-world deployment.

D. Object (Multi) Classification (14 Classes)

MIS+GPR-KNN (5-distance) emerged as standout performer, with 99.99% training accuracy of 17.28% with test accuracy, with test precision and test recall standing at 17.04% and 17.13%, respectively, resulting in performance score of 17.15% [shown in Fig. 8(a)]. GPR-DT (Entropy) showcased the training accuracy of 99.99% and the test accuracy of 48.92%, with test precision and test recall standing at 49.03% and 48.92%, respectively, yielding an overall performance score of 48.95% [shown in Fig. 8(b)]. MIS+GPR-RF (Gini) exhibited a training accuracy of 99.99% and a test accuracy of 23.49%, with test precision, test recall, and overall performance standing out at 24.35%, 23.39%, and 23.74 %, respectively [shown in Fig. 8(c)]. SVM-GPR (Linear) exhibited a training accuracy of 13.32% and a test accuracy of 14.96%, with test precision, test recall, and overall performance standing out at 15.14%, 11.72%, and 13.94%, respectively [shown in Fig. 8(d)]. MIS+GPR-MLP (50, 100) with a training accuracy of 23.01% and a test accuracy of 16.56% with test precision, test recall, and overall performance all equals 16.56% [shown in Fig. 8(e)]. MIS-IDCNN (32, 0.1, Adam) dominated with training accuracy of 19.24% and 19.04% test accuracy with the precision, recall, and overall performance scores at 35.04%, 19.04%, and 24.37%, respectively [shown in Fig. 8(f)]. Thus, the GPR-DT (Entropy) emerges as the most dependable choice for real-world deployment in multiclassification tasks. Table II(d) summarizes the performance ranking of the models with respect to each sensor for multiple classifications.

E. Feature Types in Model Training and Model Complexity Relationships

Local features are given as follows:

- characteristics within a specific region or segment of a dataset;
- providing detailed insights into localized patterns.

Global features encompass the entire dataset, capturing overall trends and relationships.

Local features, such as the mean, standard deviation, and skewness, are observed in the statistics domain, time

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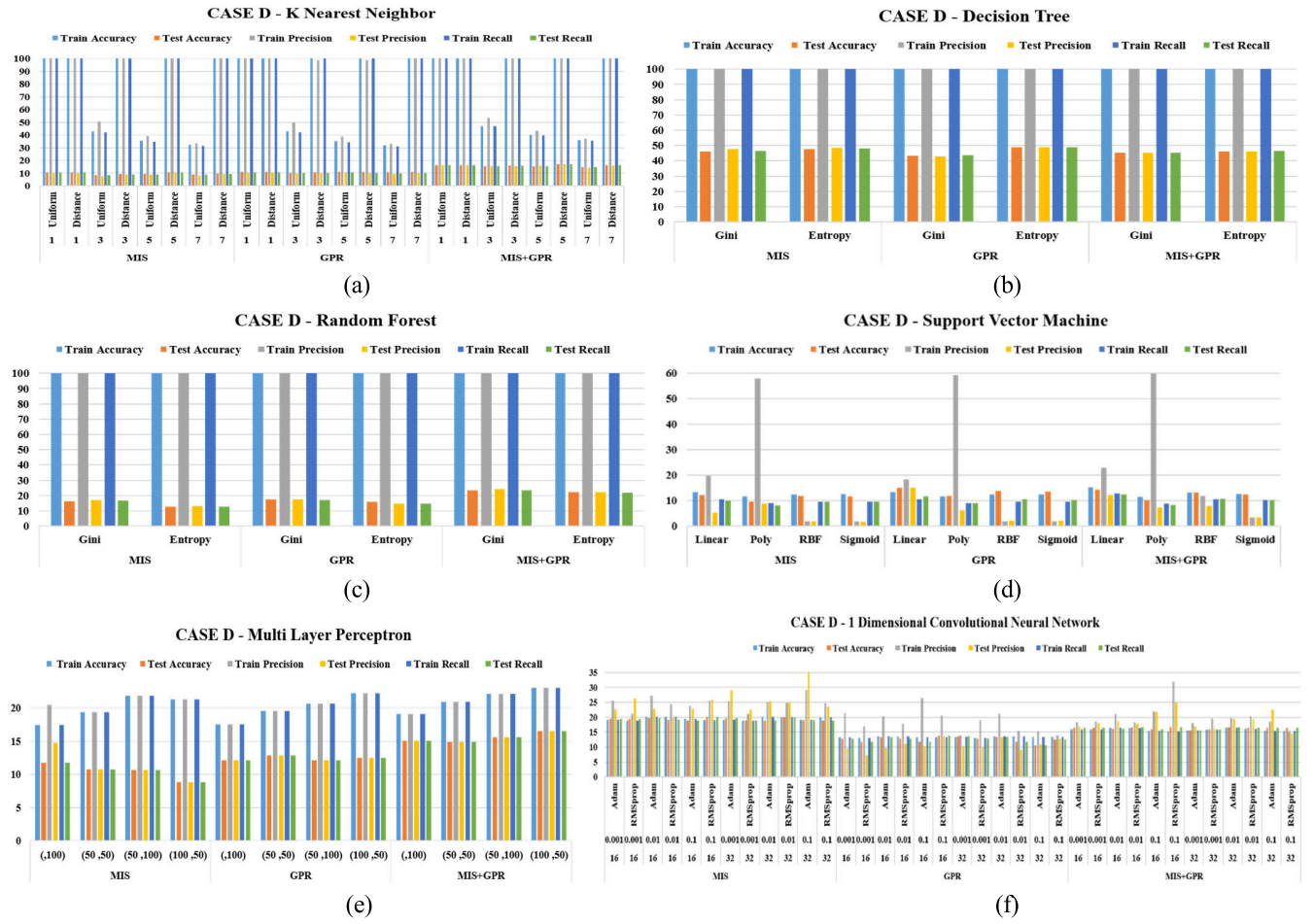


Fig. 8. Multiclassification results. (a) KNN. (b) Decision tree. (c) Random forest. (d) SVM. (e) MLP. (f) IDCNN.

domain, Hjorth parameters, and waveform/pulse characteristics. Global features, including signal duration, bandwidth, and time-domain zero crossings, are prominent in the peak and crossing domains. In model training, both features are crucial for obtaining a comprehensive understanding of the data, allowing the model to learn specific nuances and general patterns. Table III and (5)–(7) show the relationships between model complexity, weight, k, criterion for DTs and RF (Gini and Entropy), kernel type for SVM (Linear, Poly, RBF, and Sigmoid), hidden layer size for MLP, and for 1DCNN hyperparameters (BS, LR, and Ops)

$$\text{Model Complexity} \propto k \tag{5}$$

$$\text{Model Complexity} \propto \text{Criterion} \tag{6}$$

$$\text{Model Complexity} \propto \text{Kernel Type} \tag{7}$$

$$\text{Model Complexity} \propto \text{Hidden Layer Size}. \tag{8}$$

F. Literature Comparison

This study exhibits comparable results to those presented in Table IV, highlighting the effectiveness of both individual sensor and sensor fusion with the machine and deep learning methodologies employed. Normally, training accuracy is found to be much higher relative to testing accuracy. However, testing accuracy is more important as the model's behavior is observed on unseen data, which truly represents the model's performance. This is crucial for deployment in real tasks. Therefore, in this study, we have concentrated on testing results compared to the very high training results.

G. Limitations and Implications

In terms of implications, the results of this study offer insights that could inform the development of robust and reliable detection systems for various applications, including humanitarian demining, infrastructure inspection, and security operations. However, it is essential to acknowledge certain limitations of the proposed study. The performance of AI models can be affected by factors, such as hyperparameters, feature selection, and training protocols. This study emphasizes the importance of careful experimentation and validation to ensure reliable results. AI models learn patterns and features specific to the dataset during training by optimizing their parameters to minimize errors. However, when presented with unseen data, these models may need help to generalize effectively, as the unseen dataset may contain variability and characteristics not adequately represented in the training data.

Consequently, the model's performance on unseen data may degrade, resulting in lower classification accuracy or increased false-positive rates, particularly in multiobject classification

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TABLE III
MODEL COMPARISON SUMMARY FOR VARIOUS PARAMETERS IN DIFFERENT ALGORITHMS. (a) KNN, (b) DT AND RF, (c) SVM, (d) MLP, AND (e) IDCNN

(a)				(b)				(c)		
Weight	***k***	**Training Time**	**Features Required**	**Criterion**	**Number of Trees**	**Training Time**	**Features Required**	**Kernel**	**Training Time**	**Features Required**
Uniform	Small	Low	Local	**Gini**	Small	Low	Local + Global	**Linear**	Low	Local+ Global
Uniform	Large	High	Local + Global	**Gini**	Large	High	Local + Global	**Poly**	High	Local+ Global
Distance	Small	Low	Local	**Entropy**	Small	Low	Local + Global	**RBF**	High	Local+ Global
Distance	Large	High	Local + Global	**Entropy**	Large	High	Local + Global	**Sigmoid**	Moderate	Local+ Global

(d)			(e)			
Hidden Layer Size	**Training Time**	**Features Required**	**Parameter**	**Value**	**Training Time**	**Features Required**
(100)	Low	Local +Global	**Batch Size**	16	Low	Local + Global
				32	High	Local + Global
(50, 50)	High	Local +Global	**Learning Rate**	0.001	Low	Local + Global
				0.01	Balanced	Local + Global
(100, 50)	High	Local +Global		0.1	High	Local + Global
			Optimizer	Adam	Balanced	Local + Global
(50, 100)	Moderate	Local +Global		RMSprop	Balanced	Local + Global

TABLE IV
LITERATURE COMPARISON

Paper	Sensor	AI Method	Accuracy
[24]	MIS, GPR	Linear-regression model	99%
[45]	GPR	Wavelet Packet Analysis, Higher-Order Statistics	90%
[46]	GPR	CNN	99%
[47]	GPR B-scans	CNN	95%
[48]	GPR	Hidden Markov Models	90%
[49]	Geographic Information System	SVM, RF, XGBOOST	97.15%
[50]	Electromagnetic Induction	1DCNN, 2DCNN	90%
Proposed Study	MIS, GPR, MIS+GPR	KNN, DT, RF, SVM, MLP, IDCNN	99%

cases. Moreover, the deployment environment, such as edge devices with constrained memory and processing capabilities, further exacerbates this challenge. Limited computational resources may restrict the model's ability to adapt to unseen data or perform complex inference tasks, leading to suboptimal performance in real-world settings. Therefore, while the reported accuracy of AI models based on training data may appear high, it is essential to temper expectations regarding their performance on unseen datasets in practical deployment scenarios. This highlights the need for ongoing research to address these challenges and improve the robustness of AI-based detection systems in real-world applications.

IV. CONCLUSION

The present research on multi-sensor fusion for metallic and non-metallic object classification in complex soil environments and at multiple depths is a significant contribution to the field. The thorough analysis and comparison of 900 variants across six models offer valuable insights, highlighting the efficacy of the feature fusion approach (MIS+GPR) for binary, soil, depth, and multiclass classifications. Based on the provided results, the most challenging type of classification is multi-classification, involving 14 classes. Achieving high accuracy and precision in distinguishing between 14 different classes of objects is more demanding than binary classification (metallic versus non-metallic), depth classification (5 cm, 10 cm, and flush), or soil classification (cooperative, uncooperative, and rocky). The multiclassification task introduces increased complexity and variability, requiring models to differentiate among several distinct classes. IDCNN-MIS (32, 0.01, RMSprop) and (32, 0.1, Adam), and IDCNN-MIS+GPR (16, 0.1, Adam), (32, 0.1, Adam), (16, 0.01, RMSprop), and (32, 0.01, RMSprop) have emerged as the most promising model for real-time metallic versus non-metallic object classification on unseen data. The DT with the Gini criterion in the MIS+GPR scenario stood out as the optimal choice for real-time classification on an unseen soil classification dataset. IDCNN (MIS, GPR, and MIS+GPR) emerged as the best-performing model, showcasing the highest test accuracy and performance metrics, thereby solidifying its position as the most reliable choice for real-world deployment. For multiobject classification, as it was a difficult task, DT-GPR with entropy criterion performed better than other models.

The global applications of this research are extensive, ranging from defense and archeology to geology. Enhancing object detection in complex soil environments holds promising avenues for real-world scenarios. The results suggest practical solutions for landmine detection, soil classification, and depth estimation, all impacting industries worldwide. Future work could expand the dataset to include a more extensive

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range of objects and environmental conditions, improving the models' robustness. Exploring the integration of other sensor modalities beyond MIS and GPR for even more comprehensive detection capabilities is another avenue for future research. This research lays a solid foundation for object detection and classification advancements through remote sensing.

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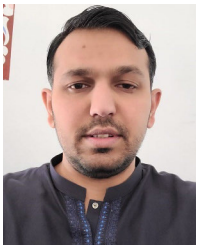
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